# Advanced Microbiology Syllabus

**BIO261** • Bella Turner

Life Sciences • Biology

## 1. Learning Objectives & Taxonomy

| ID | Learning Objective | Bloom's Level |
|---|---|---|
| LO1 | Analyze microbial cell architecture and regulatory mechanisms to explain phenotypic adaptation. | Analyze |
| LO2 | Evaluate the metabolic and genetic strategies that enable microbial survival in extreme environments. | Evaluate |
| LO3 | Synthesize knowledge of host-pathogen interactions and antimicrobial resistance to propose public health strategies. | Create |
| LO4 | Interpret complex metagenomic datasets to assess microbial community structure and function. | Analyze |
| LO5 | Apply principles of synthetic biology and biotechnology to design solutions for industrial or clinical challenges. | Apply |

**Overall Course Cognitive Depth:** Analysis and Evaluation

## 2. Weekly Instructional Plan

| Wk | Subject and Activities | Outcomes |
|---|---|---|
| 1 | **Orientation to Advanced Microbiology: scope, expectations, and contemporary research directions**<br>Syllabus • Ice Breakers | — |
| 2 | **Microbial cell architecture, genome organization, and advanced regulatory mechanisms**<br>Face to Face Lectures, Lecture Notes • Concept Maps | LO1 |
| 3 | **Microbial metabolism, energy conservation, and metabolic diversity in extreme and variable environments**<br>Asynchronous Lecture Videos • Think-Pair-Share | LO2 |
| 4 | **Microbial genetics, mutation, recombination, and horizontal gene transfer**<br>Debate / Discussion Sessions • Jigsaw • Quizzes | LO1 |
| 5 | **Microbiomes, symbiosis, and microbial ecology in host-associated and environmental systems**<br>Article Analysis • Research Report | LO4 |
| 6 | **Host-pathogen interaction, virulence strategies, and immune evasion**<br>Case Study • Socratic Seminar | LO3 |
| 7 | **Antimicrobial agents, resistance mechanisms, and stewardship**<br>Invited Speakers • Panel Discussions | LO3 |
| 8 | **Mid-semester synthesis: microbial systems integration and applied problem solving**<br>Collaborative Study Sessions • One Minute Paper • Midterm Exam | — |
| 9 | **Microbial ecology in extreme environments and biogeochemical cycling**<br>Real-World Cases • Brainstorming | LO2 |
| 10 | **Metagenomics, microbial community profiling, and bioinformatic interpretation**<br>Area Specific Software • Live Coding | LO4 |
| 11 | **Advanced virology: replication strategies, host specificity, and viral evolution**<br>Peer Instruction • Seminars | LO1 |
| 12 | **Microbial biotechnology, industrial applications, and synthetic biology**<br>Workshops • Project pitch | LO5 |
| 13 | **Microbial diagnostics, surveillance, and public health microbiology**<br>Learning Management System • Discussion Forums • Assignments / Homework | LO3, LO5 |

| Wk | Subject and Activities | Outcomes |
|---|---|---|
| 14 | **Emerging infectious diseases, outbreak investigation, and microbial risk communication**<br>Simulations • Breakout Rooms | LO3 |
| 15 | **Student-led synthesis, review of advanced themes, and exam preparation**<br>Practice Exams • Exit Tickets | — |
| 16 | **Final Exam**<br>Final Exam | — |

## 3. Skill Acquisitions by Category

| Domain | Specific Skill | Related Wk |
|---|---|---|
| **Hard** | Metabolic Pathway Interpretation | 3, 9 |
| | Bioinformatic Data Analysis | 10 |
| **Soft** | Scientific Literature Synthesis | 1, 5, 6 |
| | Risk Communication | 14 |
| **Meta** | Systems Thinking | 8, 15 |
| | Evidence-Based Decision Making | 13, 14 |

## 4. Proposed Micro-Credential Framework

| Credential Name | Dimensions | Depth | Align. | Mapping |
|---|---|---|---|---|
| Applied Microbial Genomics Specialist | Application | Proficiency | W4, 10 | #BioinformaticDataAnalysis #MicrobialGenetics |
| Public Health Microbiology Strategist | Integration | Mastery | W7, 13, 14 | #RiskCommunication #EvidenceBasedDecisionMaking |

## 5. Assessment & Mapping

| Activity Name | Week | Weight | Bloom's | Outcomes |
|---|---|---|---|---|
| Quizzes | 4 | 15% | Understand | LO1, LO2 |
| Research Report | 5 | 20% | Analyze | LO4 |
| Midterm Exam | 8 | 30% | Analyze | LO1, LO2, LO3 |
| Final Exam | 16 | 35% | Evaluate | LO1, LO2, LO3, LO4, LO5 |